Connector Economies
变量构造目录与使用指南

内部严格审查材料

variable_catalog_v0.1.1

由 F:/working_paper_wenxiao/trade_welfare/docs/variable_construction_catalog.md
可重复生成

用途：合作者复核数据来源、处理规则、变量定义、验证结果和解释边界。

目录

Material Passport 4
0. 结论和使用方式 4
1. 状态、粒度和可加总性代码 4
1.1 状态 . . . 4
1.2 粒度代码 . . . 5
1.3 可加总性代码 . . . 5
2. 通用键、标签和 provenance 变量 5
3. M2 已验证的政策变量 6
4. M2 已验证的贸易金额和 Census 变量 7
4.1 Connector trade . . . 7
4.2 Census . . . 7
5. M3.1 connector-sector-year 基础聚合变量 7
5.1 金额和覆盖 . . . 7
5.2 sector-level treatment . . . 8
6. M3.2 时间窗口变量 8
6.1 zero 和 small-base flags . . . 9
7. M3.3-M3.6 贸易重组和排名输入变量 9
7.1 Import replacement . . . 9
7.2 Source-to-connector input growth . . . 10
7.3 Product-level co-movement . . . 10
7.4 Exposure overlap . . . 10
7.5 稳定性和规模 . . . 10
8. M4 production-side 标准化变量 10
8.1 通用 long schema . . . 11
8.2 ICIO metrics . . . 11
8.3 AMNE metrics . . . 11
8.4 BEA MNE metrics . . . 12
8.5 BEA FDI metrics . . . 12
8.6 Mexico FDI metrics . . . 12
9. M4-M5 production changes 和覆盖变量 12
10. M6 connector ranking 13
10.1 组件 . . . 13
10.2 至少两种排名方案 . . . 13
10.3 Ranking outputs . . . 14
11. M6 rerouting facts 和 production validation 14
11.1 Rerouting classification . . . 14

11.2 Production validation 14
12. Structural moments 和 Scenario A/B/C 15
12.1 Trade-cost candidates 15
12.2 MP/production targets 15
12.3 Structural moment table 15
13. Controls、fixed effects 和 robustness 变量 16
14. 不允许的变量和解释 16
15. Stata 文件、主键和验证合同 17
15.1 计划交付 17
15.2 Storage types 17
15.3 必须运行的 Stata 检查 17
16. 每个变量完成后的登记模板 18
17. 当前施工顺序 18
18. 版本记录 18

Material Passport

- Origin Skill: stata-latex-research
- Origin Date: 2026-07-27
- Verification Status: DESIGN_REVIEWED
- Version Label: variable_catalog_v0.1.1
- Current Data Stage: M0-M2 completed
- First Construction Stage: M3.1 not started
- Canonical Format: Parquet
- Collaborator Format: Stata .dta

0. 结论和使用方式

从 M3.1 开始，变量构造应以本目录为变量合同。它回答六个问题：

1. 变量在哪一层数据上构造；
2. 变量的公式、分母、时间窗口和单位是什么；
3. 变量用于 connector ranking、rerouting、production validation 还是 structural moments；
4. 零值、缺失、负值、small base、suppression 和 confidentiality 如何处理；
5. 变量能否求和、能否跨 connector 比较、能否进入回归或模型；
6. 导出 Stata 时使用什么名称、storage type、主键和验证命令。

本目录同时包含三类变量：

- EXISTING_VALIDATED：M2 已存在且 validation PASS；
- PLANNED_M3 至 PLANNED_M6：已经定义但尚未构造，必须在对应阶段完成后补充实际 output hash 和 validation；
- STRUCTURAL_TARGET_ONLY：只能作为模型校准目标或输入候选，不能直接解释为已经识别的结构参数。

凡是本目录未登记的正式变量，不应直接写入最终 processed data。临时探索变量可以存在于 tmp/ 或交互式 Stata session，但进入正式输出前必须补齐本目录。

本目录与 docs/data_processing_audit_trail.md 构成同一套审查材料。本目录不重复数据来源和完整处理历史，而是引用审计档案中的输入、输出和 validation 证据；审计档案也不能替代这里的变量级公式、用途和解释边界。M3-M7 任一阶段完成时，两份 Markdown、两份 PDF 和对应 build summary 必须同步更新。

1. 状态、粒度和可加总性代码

1.1 状态

状态	含义
EXISTING_VALIDATED	已在 processed Parquet 中存在，主键和 validation 已通过
PLANNED_M3	connector-product-year 聚合到 connector-sector-year 时构造
PLANNED_M4	production-side standardized long table 时构造
PLANNED_M5	trade 和 production evidence 合并时构造
PLANNED_M6	ranking、rerouting、validation 和 structural moments 时构造
STRUCTURAL_TARGET_ONLY	观测 moment 或 candidate shock，不等于已识别结构参数
PROHIBITED	明确禁止构造或禁止解释的变量

1.2 粒度代码

Grain ID	唯一键	主要文件
G_PRODUCT_YEAR	analysis_hs6, year	product treatment
G_CONN_PRODUCT_YEAR	connector_iso3, analysis_hs6, year	connector trade
G_ORIGIN_PRODUCT_MONTH	origin_iso3, analysis_hs6, year, month	Census monthly
G_CONN_SECTOR_YEAR	connector_iso3, analysis_sector, year	M3 trade-sector
G_CONN_SECTOR_WINDOW	connector_iso3, analysis_sector, window_id	M3 window summaries
G_PROD_LONG	country_iso3, analysis_sector, year, source_name, metric_name，必要时增加 direction/parent	M4 production long
G_EVIDENCE	connector_iso3, analysis_sector, year	M5 integrated evidence
G_RANK	connector_iso3, analysis_sector, ranking_spec	M6 ranking
G_MOMENT	moment_id, scenario	M6 structural moments

1.3 可加总性代码

Code	规则
ADD_WITHIN_KEYS	可以在不跨越声明维度时求和，例如产品金额聚合到 sector
NONADDITIVE_RATE	rate/share/exposure 不能直接求和，必须重算分子分母
NONADDITIVE_REPEAT	为比较而重复，绝不能跨重复维度求和
STOCK	时点/期末 stock，不能跨年求和
FLOW	期间 flow，可在不重叠期间内求和
LEVEL	level，可比较或求差，但通常不跨单位/来源相加
FLAG	逻辑状态，不作金额加总

2. 通用键、标签和 provenance 变量

这些字段必须尽可能贯穿 M3-M7。它们不是辅助杂项，而是合作者复核 join 和解释边界的基础。

Canonical name	Stata name	Type	Grain	用途	状态
connector_iso3	conn_iso3	str3	connector grains	connector 国家键	existing/planned
country_iso3	country_iso3	str3	production	production host/country 键	planned M4
analysis_hs6	hs6	str6	product grains	保留 leading zeros 的产品键	existing
analysis_sector	ana_sector	str32	sector grains	第一阶段共同经验 sector	existing
year	year	int	annual grains	calendar year	existing
month	month	byte	monthly	1-12	existing
window_id	window_id	str8	window	pre/shock/post	planned M3
source_name	source	str20	production long	数据来源区分	planned M4
metric_name	metric	str32	production long	指标区分	planned M4
unit	unit	str32	all long outputs	单位，禁止跨单位相加	planned M4
mapping_version	map_ver	str32	all	concordance 版本	planned
build_version	build_ver	str32	all	builder/output schema 版本	existing/planned
source_row_count	src_n	long	all	聚合 cell 的原始行数	existing/planned
source_file_count	srcfile_n	int	all	provenance 完整性	existing/planned
mapping_status	map_status	str32	all	mapped/aggregate/residual/ambiguous 等	existing/planned
is_analysis_eligible	ana_ok	byte	all	是否进入第一阶段主分析	existing/planned
is_proxy	is_proxy	byte	all	TWN S19 或其他 proxy	existing/planned
is_aggregate	is_agg	byte	production/country	aggregate geography/sector	existing/planned
is_residual	is_resid	byte	country	Mexico residual 等	existing/planned

Canonical name	Stata name	Type	Grain	用途	状态
suppressed_flag	suppressed	byte	production	BEA suppressed	planned M4
confidential_flag	confidentl	byte	production	Mexico confidential	existing/planned
allocated_flag	allocated	byte	production	AMNE allocated metric	planned M4
nonadditive_flag	nonadd	byte	all	repeated/rate/stock 不可加总	planned

Stata 中所有逻辑变量使用 numeric byte 0/1，并附 value label。不能把 Python boolean 转成字符串 "True"/"False"。

3. M2 已验证的政策变量

文件：data/processed/harmonized/product_treatment_hs6_year.parquet。粒度 G_PRODUCT_YEAR。

Canonical name	Stata name	定义/单位	可加总性	用途	状态
total_us_import_value_2017	usimp17	2017 total U.S. imports，USD	level；映射后可加总	treatment 分母	existing validated
china_import_value_2017	chnimp17	2017 China imports，USD	level；映射后可加总	China share 和 weights	existing validated
china_import_share_2017	chnshr17	chnimp17/usimp17，fraction	nonadditive rate	exposure 分母结构	existing validated
tariff_rate_china_weighted	trf_chw	source HS17 rate 按 China imports 加权	nonadditive rate	sensitivity/provenance	existing validated
tariff_rate_total_import_weighted	trf_usw	source rate 按 total imports 加权	nonadditive rate	zero-China fallback	existing validated
tariff_rate	trf_rate	最终 statutory rate，fraction	nonadditive rate	policy intensity	existing validated
tariff_exposure	trf_expo	sum(China imports × rate)/sum(total imports)	nonadditive rate	主 gross exposure	existing validated
tariffed_product	tariffed	任一 mapped source code rate > 0	flag	treatment indicator	existing validated
source_hs17_code_count	hs17_n	汇入 analysis product 的 HS17 数量	count	audit many-to-one	existing validated
policy_source_observed_count	polobs_n	legal panel 中 observed source codes	count	区分 not-listed 与 missing	existing validated
source_tariffed_hs17_count	trfhs17_n	positive-rate source codes	count	treatment composition	existing validated
tariff_rate_max_source	trf_max	mapped sources 最大法定 rate	nonadditive rate	denominator=0 fallback	existing validated
policy_mapping_status	pol_map	weighted/no denominator/no HS17 weight	category	sample/filter/audit	existing validated
historical_exclusions_applied	hist_excl	当前恒为 0	flag	防止误称 realized duties	existing validated
four_year_review_2024_included	rev24_in	当前恒为 0	flag	防止误并 2024 review	existing validated

禁止变量：

Prohibited name/concept	原因
realized_tariff_paid from current panel	当前数据未重建 customs-entry realized duties
tariff_exposure_with_exclusions	historical exclusions 尚未应用
2024_review_hs6 by first-six-digit truncation	modern HTS scope 和 historical HS6 不可直接等同

4. M2 已验证的贸易金额和 Census 变量

4.1 Connector trade

文件：connector_trade_hs6_year.parquet。粒度 G_CONN_PRODUCT_YEAR。

Canonical name	Stata name	单位	可加总性	用途	状态
china_to_usa_value_usd	cn_us_val	current USD	NONADDITIVE_REPEAT across connector	共同 China-USA benchmark	existing validated
china_to_connector_value_usd	cn_conn_val	current USD	add across products, not connectors unless desired	China input leg	existing validated
east_asia_to_connector_value_usd	ea_conn_val	current USD	add across products	East Asia input leg	existing validated
connector_to_usa_value_usd	conn_us_val	current USD	add across products/connectors with clear scope	connector export leg	existing validated
*_row_count	shortened *_n	count	additive count	区分 true zero 与 source rows	existing validated
east_asia_self_trade_excluded	ea_selfx	flag	flag	KOR/TWN self-trade rule	existing validated
baci_absence_treated_as_zero	baci_zrule	flag	flag	complete-grid zero rule	existing validated
china_to_usa_common_benchmark	cnus_common	flag	flag	repeated warning	existing validated
treatment_repeated_across_connectors	trf_repeat	flag	flag	non-additive warning	existing validated

4.2 Census

月度 grain G_ORIGIN_PRODUCT_MONTH；年度 grain origin_iso3, analysis_hs6, year。

Canonical name	Stata name	单位/公式	用途	状态
import_value_usd	imp_val	current USD	主进口金额	existing validated
customs_value	cust_val	current USD	customs valuation check	existing validated
duties	duties	current USD	collected duties 描述，不等于 Section 301-only duties	existing validated
dutiable_value_usd	dutiable	current USD	dutiable base	existing validated
observed_month_count	month_n	0-12	年度 cell 的月度覆盖	existing validated
calendar_source_months_complete	cal12_ok	flag	允许 sparse absence -> zero 的前提	existing validated
quantity_usable	qty_ok	当前恒为 0	禁止 quantity analysis	existing validated
is_all_origin_denominator	allorig	当前恒为 0	防止误作全来源分母	existing validated

duties/import_value 可以作为描述性 average duty ratio，但不能命名为 Section 301 realized rate，除非另行识别 tariff program 和 entry-level scope。

5. M3.1 connector-sector-year 基础聚合变量

目标文件：data/processed/harmonized/connector_trade_sector_year.parquet。粒度 G_CONN_SECTOR_YEAR。

5.1 金额和覆盖

对 is_analysis_sector_eligible=true 且 product-sector weight=1 的产品，在 connector_iso3, analysis_sector, year 内求和。

Canonical name	Stata name	公式	可加总性	用途	状态
china_to_usa_value_usd	cn_us_val	sum_hs6(cn_us_val)，但每个 connector 重复	nonadditive across connector	China decline benchmark	planned M3
china_to_connector_value_usd	cn_conn_val	sum_hs6(cn_conn_val)	flow	China input growth	planned M3
east_asia_to_connector_value_usd	ea_conn_val	sum_hs6(ea_conn_val)	flow	broader source input growth	planned M3
connector_to_usa_value_usd	conn_us_val	sum_hs6(conn_us_val)	flow	connector export growth	planned M3
included_product_count	prod_n	eligible HS6 count	count	sector composition	planned M3
positive_trade_product_count	posprod_n	amount > 0 的 HS6 count	count	extensive product margin	planned M3
excluded_value_usd	excl_val	同 connector-year 中 excluded/outside/residual products value	flow	披露样本缩窄	planned M3
included_value_share	incl_share	included / (included + excluded)	nonadditive share	sector coverage	planned M3

每条金额字段分别计算 source_row_count。不能只保留金额而丢失 source-row diagnostics。

5.2 sector-level treatment

Policy rate/exposure 不能按产品直接求和。推荐保留多个定义：

Canonical name	Stata name	公式	用途	状态
sector_tariff_rate_china_weighted	sec_trf_cw	sum_h(chnimp17_h × rate_h,t)/sum_h(chnimp17_h)	法定 rate intensity	planned M3
sector_tariff_exposure	sec_expo	sum_h(chnimp17_h × rate_h,t)/sum_h(usimp17_h)	主 policy exposure	planned M3
sector_china_import_share_2017	sec_cnshr	sum_h(chnimp17_h)/sum_h(usimp17_h)	baseline China dependence	planned M3
tariffed_product_share_count	trf_psh_n	tariffed HS6 / eligible HS6	breadth	planned M3
tariffed_product_share_china_value	trf_psh_c	tariffed China imports / sector China imports	treatment mass	planned M3
tariffed_product_share_us_value	trf_psh_u	tariffed total imports / sector total imports	denominator exposure breadth	planned M3
treatment_missing_denominator_share	trf_miss	no-denominator products 的 baseline/value share	audit	planned M3

主回归/排名应优先使用 sector_tariff_exposure，同时把 weighted rate 和 tariffed shares 作为 robustness，不把它们互相替代。

6. M3.2 时间窗口变量

不同窗口长度不相等：pre 4 年、shock 2 年、post 5 年。因此 level comparison 使用年度平均值，不使用窗口总和。

对任意非负年度 flow x_t：

```
x_pre  = mean(x_t, 2014-2017)
x_shk  = mean(x_t, 2018-2019)
x_post = mean(x_t, 2020-2024)

d_x_post_pre = x_post - x_pre
g_dh_x_post_pre = 2 * (x_post - x_pre) / (x_post + x_pre)
g_log_x_post_pre = ln(x_post) - ln(x_pre), only if both > 0
```

g_dh 是推荐 primary growth：无量纲、允许一端为 0、范围 [-2,2]。g_log 只用于 positive-base robustness。禁止把 undefined log growth 自动设为 0。

每条贸易腿构造：

Family	Pre level	Shock level	Post level	Level change	DH growth	Positive-base log growth
China -> USA	cnus_pre	cnus_shk	cnus_post	d_cnus	gdh_cnus	gln_cnus
China -> connector	cncn_pre	cncn_shk	cncn_post	d_cncn	gdh_cncn	gln_cncn
East Asia -> connector	eacn_pre	eacn_shk	eacn_post	d_eacn	gdh_eacn	gln_eacn
connector -> USA	cnusx_pre	cnusx_shk	cnusx_post	d_cnusx	gdh_cnusx	gln_cnusx

Stata 名称必须在生成前再次检查不超过 32 characters，并在 variable labels 写出完整方向。

6.1 zero 和 small-base flags

Canonical name	Stata name	定义	用途
pre_zero_flag	pre_zero	pre annual mean = 0	区分新贸易与连续增长
post_zero_flag	post_zero	post annual mean = 0	区分退出
small_base_flag	smallbase	pre level 低于 M3 config 明确阈值	防止 ratio/rank 爆炸
log_growth_defined	logg_ok	pre>0 and post>0	log sample
years_observed_pre	pre_yn	pre 窗口实际 years	coverage
years_observed_post	post_yn	post 窗口实际 years	coverage

small-base threshold 必须写入 M3 config，至少报告：

- absolute USD threshold；
- sector-relative percentile threshold；
- 不使用 threshold、只用 DH growth 的结果。

推荐主排名使用 level change + DH growth，不依赖单一百分比增长。

7. M3.3-M3.6 贸易重组和排名输入变量

7.1 Import replacement

```
china_us_decline = max(cnus_pre - cnus_post, 0)
connector_us_gain = cnusx_post - cnusx_pre
replacement_ratio = connector_us_gain / china_us_decline
```

replacement_ratio 只有在 china_us_decline 大于明确的 denominator threshold 时定义；否则 missing。主结果同时保留 level gain 和 decline，不能只报告 ratio。

Canonical name	Stata name	用途	状态
china_us_decline_usd	cnus_drop	China displacement level	planned M3
connector_us_gain_usd	conn_gain	connector gain level	planned M3
replacement_ratio	repl_ratio	connector gain 相对 China decline	planned M3
replacement_ratio_defined	repl_ok	denominator 是否可用	planned M3
replacement_capped_for_rank	repl_cap	仅 ranking 的 winsorized/capped version	planned M3

7.2 Source-to-connector input growth

Canonical name	Stata name	定义
china_input_gain_usd	cnin_gain	China -> connector post-pre level change
east_asia_input_gain_usd	eain_gain	East Asia -> connector post-pre level change
china_input_dh_growth	gdh_cnin	China input DH growth
east_asia_input_dh_growth	gdh_eain	East Asia input DH growth
source_export_ratio_pre	srcx_pre	source-to-connector / connector-to-USA，pre
source_export_ratio_post	srcx_post	同一 ratio，post
source_export_ratio_change	d_srcx	post-pre

ratio denominator 为 0 时保持 missing；不能用 arbitrary epsilon 生成正式值。若 ranking 需要 finite score，先用 percentile of level/growth components，而不是偷偷给 ratio 加常数。

7.3 Product-level co-movement

在 connector-sector 内先对每个 HS6 计算 pre/post changes，再聚合：

Canonical name	Stata name	公式/用途
both_legs_positive_product_share	both_psh	China/East Asia input gain >0 且 connector-US gain >0 的 HS6 share
both_legs_positive_value_share	both_vsh	上述产品按 pre/post trade value 加权 share
china_decline_connector_gain_share	rerout_psh	China-US decline 且 connector-US gain 的 product share
triple_condition_product_share	triple_psh	China-US decline、source input gain、connector-US gain 同时成立
product_change_correlation	prod_corr	source-input change 与 connector-export change 的 HS6 correlation
weighted_product_change_correlation	wprod_cor	以 pre China-US 或 connector trade weights 加权

相关系数至少要求有效 products 达到 config 中的 minimum count；否则 missing + corr_low_n_flag=1。

7.4 Exposure overlap

Canonical name	Stata name	公式
connector_export_tariff_overlap_pre	ovlp_pre	pre connector-US exports 中 tariffed/exposed product 的 value share
connector_export_tariff_overlap_post	ovlp_post	post share
connector_export_exposure_weighted	ovlp_expo	connector export weights 下的 product tariff exposure
gain_in_high_exposure_products_share	gain_hiexp	export gain 来自 high-exposure products 的份额

“high exposure” threshold 必须在 config 中固定，例如 top quartile 或 positive exposure；两种定义均应报告。

7.5 稳定性和规模

Canonical name	Stata name	用途
pre_trade_scale_usd	scale_pre	排名时防止只由小基数增长驱动
post_trade_scale_usd	scale_post	经济规模
annual_growth_sd	growth_sd	年度增长波动
positive_growth_year_share	posyr_sh	post 期间正增长年份比例
rank_window_sensitivity	rank_sens	替代窗口下 rank 变化
source_bloc_sensitivity	bloc_sens	China-only 与 East-Asia source 定义差异
exposure_definition_sensitivity	expo_sens	rate/exposure/tariffed share 差异

8. M4 production-side 标准化变量

目标文件：production_validation_long.parquet。采用长表保存来源、指标与单位；禁止把不同概念合并相加。

8.1 通用 long schema

Canonical name	Stata name	定义
country_iso3	country_iso3	host/country
analysis_sector	ana_sector	共同 sector
year	year	年份
source_name	source	icio/amne/bea_mne/bea_fdi/mexico_fdi
metric_name	metric	指标名
value	value	原单位数值
unit	unit	current USD/current million USD/thousand persons/fraction
frequency	freq	annual/quarterly/H1
direction	direction	FDI inward/outward 或 affiliate trade direction
parent_country_iso3	parent_iso3	AMNE parent，适用时
direct_or_allocated	direct_alloc	direct/allocated/unavailable
suppressed_flag	suppressed	BEA
confidential_flag	confidentl	Mexico
aggregate_flag	is_agg	geography/sector aggregate
proxy_flag	is_proxy	coarse source-sector proxy
source_file	src_file	provenance

8.2 ICIO metrics

Metric	Stata metric label	Unit	Additivity/use
gross_output	gross_output	current million USD	production level
value_added	value_added	current million USD	production validation；允许 accounting negatives
exports	exports	current million USD	source-consistent export level
imports	imports	current million USD	source-consistent import level
intermediate_inputs	intermed_inputs	current million USD	IO intensity
taxes_less_subsidies	tax_less_sub	current million USD	accounting；允许负值
accounting_residual	acct_resid	current million USD	quality diagnostic

ICIO 只有 2017、2018、2022。禁止线性插值得到 2019 或 annual panel。

8.3 AMNE metrics

Metric	Unit	Direct/allocated	用途
affiliate_output	current million USD	direct bilateral	MP activity 主指标
affiliate_value_added	current million USD	allocated to parent	production validation
affiliate_exports	current million USD	allocated	affiliate trade
affiliate_imports	current million USD	allocated	affiliate input
foreign_affiliate_output_total	current million USD	host-sector total	foreign activity scale
parent_affiliate_output_share	fraction	derived direct output share	allocation provenance
affiliate_employment	unavailable	unavailable	PROHIBITED as zero/imputed

8.4 BEA MNE metrics

Metric	Unit	用途
employment	thousand employees	labor validation
sales	million current USD	affiliate activity
value_added	million current USD	production
capital_expenditures	million current USD	investment/activity
goods_exports_to_us	million current USD	affiliate -> U.S. direction
goods_imports_from_us	million current USD	U.S. -> affiliate direction

每个 metric 同时保留 suppression 和 small-value flag。aggregate geography 先排除。Transportation equipment 若作为 autos robustness，必须 proxy_flag=1。

8.5 BEA FDI metrics

Metric	Temporal type	Unit	规则
position	stock	million current USD	不跨年求和
flows	annual flow	million current USD	允许负 reversal
income	annual flow	million current USD	与 position/flow 分开

必须保留 direction。不能构造 total_fdi = position + flows + income。

8.6 Mexico FDI metrics

Metric	Stata name	规则
fdi_quarterly_usd	fdi_q_usd	quarterly flow，负值保留
fdi_annual_usd	fdi_a_usd	仅四季度完整且均非 confidential 时求和
fdi_h1_usd	fdi_h1_usd	2024 与历史 H1-to-H1 比较
fdi_complete_year_flag	fdi_yr_ok	2006-2023 full-year；2024 detail false
fdi_confidential_quarter_count	fdi_conf_n	confidential quarter count
fdi_disclosed_composition_share	fdi_discsh	只描述 disclosed cells，不能解释为完整 national share

Mexico national total 与 detailed composition 必须存为不同 source/metric，不能用 disclosed detail 替代全国 total。

9. M4-M5 production changes 和覆盖变量

对每个 source-metric 在其实际可用年份内构造变化，不能强迫所有来源使用同一连续窗口。

Canonical name	Stata name	公式/规则
value_level	value	原始 level
value_lag	value_lag	同 source-country-sector-metric 上一期
value_change	d_value	value-value_lag
value_dh_growth	gdh_value	非负 level 适用；signed FDI 不使用 DH level formula
value_log_growth	gln_value	两期均正时
signed_flow_change	d_flow	signed FDI flow 的 level change
post_pre_mean_change	d_postpre	在 source 可覆盖窗口内年度平均差
coverage_year_count	cover_yn	有效年份数
missing_value_count	miss_n	missing 数
suppressed_value_count	supp_n	suppressed 数
confidential_value_count	conf_n	confidential 数

M5 coverage flags:

Canonical name	Stata name
has_trade	has_trade
has_icio	has_icio
has_amne	has_amne
has_bea_mne	has_beamne
has_bea_fdi	has_beafdi
has_mexico_fdi	has_mexfdi
has_any_production	has_prod
has_direct_activity_measure	has_direct
has_only_financial_evidence	fin_only
source_count_available	source_n

has_only_financial_evidence=1 的 cell 不能标为 production confirmed。

10. M6 connector ranking

10.1 组件

推荐先把 raw components 转为同一比较组内的 percentile rank [0,1]，减少单位和极端值影响，同时永久保留 raw variables。

Component	Candidate raw inputs	经济含义
score_exposure	exposure overlap、sector tariff exposure	是否集中在受冲击产品
score_us_gain	connector-US level gain、DH growth	对美出口扩张
score_source_gain	China/East-Asia input gain、DH growth	source-to-connector link
score_replacement	China decline + connector gain、replacement ratio	U.S. import replacement
score_comovement	both/triple product share、weighted correlation	两条贸易腿同步性
score_scale	pre/post trade scale	经济规模
score_stability	positive-year share、negative volatility	时间稳定性
score_production	direct activity measures	生产侧验证

10.2 至少两种排名方案

方案 A，推荐主结果：透明 equal-weight percentile score。

```
rank_score_equal =
mean(score_exposure,
    score_us_gain,
    score_source_gain,
    score_replacement,
    score_comovement,
    score_scale,
    score_stability)
```

优点：透明、可复核、对单位不敏感。缺点：默认各维度同等重要。

方案 B，推荐稳健性：rerouting-core score，减少规模权重。

```
rank_score_rerouting =
0.20 exposure
+ 0.20 us_gain
+ 0.20 source_gain
+ 0.20 comovement
+ 0.10 replacement
+ 0.05 scale
+ 0.05 stability
```

优点：更贴近两段式 rerouting 机制。缺点：权重含研究判断，必须报告替代权重。

production evidence 不建议直接混入主 trade ranking；更清楚的做法是先得到 trade ranking，再给出 production_validation_grade。如果合成总分，必须作为第三套扩展结果。

10.3 Ranking outputs

Canonical name	Stata name	状态
rank_score_equal	score_eq	planned M6
rank_position_equal	rank_eq	planned M6
rank_score_rerouting	score_rr	planned M6
rank_position_rerouting	rank_rr	planned M6
rank_top3_equal	top3_eq	planned M6
rank_top3_rerouting	top3_rr	planned M6
rank_stability_across_specs	rank_stab	planned M6
rank_component_min	score_min	planned robustness
ranking_spec	rank_spec	planned M6

所有 rank 必须在包含 core connectors 和 comparators 的完整共同样本上计算，不能先删掉低排名观察。

11. M6 rerouting facts 和 production validation

11.1 Rerouting classification

Canonical name	Stata name	定义
china_us_decline_flag	cnus_dropf	China-US post < pre，且 level decline 达阈值
connector_us_gain_flag	conn_gainf	connector-US post > pre
source_input_gain_flag	src_gainf	China/East Asia -> connector post > pre
rerouting_two_leg_flag	rr_2leg	connector gain + source input gain
rerouting_three_condition_flag	rr_3cond	China decline + connector gain + source input gain
high_exposure_rerouting_flag	rr_hiexp	triple condition + high exposure
rerouting_evidence_grade	rr_grade	none/weak/moderate/strong，规则写入 config

Evidence grade 必须由预先固定的规则生成，不能根据结果手工改标签。

11.2 Production validation

Canonical name	Stata name	定义
output_growth_positive	out_pos	ICIO/direct output change >0
value_added_growth_positive	va_pos	VA change >0
affiliate_activity_growth_positive	aff_pos	AMNE/BEA activity >0
employment_growth_positive	emp_pos	available employment >0
capex_growth_positive	capex_pos	BEA capex >0
fdi_flow_increase	fdi_pos	signed flow change >0
direct_activity_signal_count	direct_n	output/VA/affiliate/employment/capex positive count
financial_signal_count	fin_n	FDI flow/position/income signals
production_confirmed_flag	prod_conf	至少一个 direct activity signal，不能只靠 FDI
financial_support_only_flag	fin_supp	只有 financial evidence
production_validation_grade	prod_grade	none/financial-only/partial/multiple-source

推荐 grade：

- none：无生产或金融支持；
- financial_only：只有 FDI；

- partial_direct：至少一个 direct activity metric 同向；
- multiple_direct：至少两个不同 source 或不同 direct activity families 同向；
- conflicting：trade 与 production signals 方向冲突。

12. Structural moments 和 Scenario A/B/C

必须区分“观测 moment”和“model shock/parameter”。数据不会直接观测 hgamma。

12.1 Trade-cost candidates

Canonical name	Stata name	公式	状态
gross_tariff_factor_hat	htau_gross	(1+tariff_rate_post)/(1+tariff_rate_pre)	structural target only
exposure_factor_hat	htau_expo	(1+exposure_post)/(1+exposure_pre)	robustness candidate
tariff_rate_change_pp	d_trf_pp	100*(rate_post-rate_pre)	descriptive moment
tariff_exposure_change_pp	d_exp_pp	100*(expo_post-expo_pre)	descriptive moment

最终 h_tau 选择必须与模型中 iceberg trade cost 或 tariff wedge 的定义一致，不能仅因为变量名相似就直接导入。

12.2 MP/production targets

Moment	观测来源	用途
foreign affiliate output share change	AMNE	incumbent MP-intensive adjustment
affiliate output growth	AMNE	Scenario B target
affiliate sales/VA/employment/capex growth	BEA MNE	direct activity validation
gross output/value added growth	ICIO	broader production response
FDI position/flow change	BEA/Mexico	Scenario C financial upper-bound support

Candidate names：

- mp_affiliate_output_share_change -> d_affshr
- mp_affiliate_output_dh_growth -> gdh_affout
- bea_affiliate_activity_index -> bea_affidx
- production_activity_index -> prod_idx
- fdi_financial_index -> fdi_idx

这些是 moments，不是 hgamma 本身。模型校准后才生成：

- hgamma_scenario_b
- hgamma_scenario_c

并必须记录 calibration mapping、normalization、elasticity、target source 和 solver result。

12.3 Structural moment table

目标字段：

Field	含义
moment_id	稳定唯一标识
scenario	A/B/C
country_iso3	country/connector
analysis_sector	sector
moment_name	名称
moment_value	数值
unit	单位

Field	含义
source_name	来源
sample_start, sample_end	样本期
formula_version	公式版本
direct_or_proxy	direct/proxy/allocated/financial
can_enter_model	是否允许进入模型
exclusion_reason	不允许时原因
validation_status	PASS/FAIL

13. Controls、fixed effects 和 robustness 变量

这些变量用于后续实证，不应与核心 outcome 混合。

Family	Variables
time	year、post2018、post2020、pre/shock/post indicators
connector FE	connector_iso3 categorical
sector FE	analysis_sector categorical
baseline scale	log pre trade level only when positive；pre level raw
baseline China dependence	sector China share 2017
baseline exposure	2017/2018 sector exposure
proxy/scope	TWN proxy、BEA autos proxy、medical unsupported
coverage	has source flags、missing/suppression/confidential counts
small base	pre_zero、small_base、log_defined
alternative windows	2014-2017 vs 2020-2023；exclude 2024；shock-only
source bloc	China-only vs extended East Asia
policy	rate vs exposure vs tariffed share

2024 可能同时受政策 review 和数据时间边界影响。所有包含 2024 的结果要有 exclude-2024 robustness。

14. 不允许的变量和解释

禁止项	原因
把 Census 15-origin 合计命名为 world imports	分母范围错误
把 Census quantity zero 当真实数量	source unavailable
把 suppression/confidential missing 填 0	改变官方含义
把 BEA position + flow + income 相加	stock/flow/概念不同
把 FDI increase 命名为 plant entry	financial evidence 不能证明实体进入
把 trade rerouting flag 命名为 production relocation	缺少 direct production evidence
把 allocated AMNE VA/exports/imports 称为直接 bilateral observations	来源口径错误
把 repeated China-US benchmark 跨 connector 求和	机械重复
把 rate/share/exposure 直接求和到 sector	应重算分子分母
给 many-to-many concordance 等权拆分且不披露	无经验依据且会伪精细
用 arbitrary epsilon 让 ratio/log 看似完整	隐藏 undefined denominator
将 Scenario C 称为 structurally identified entry margin	当前模型不是 endogenous fixed-cost entry model

15. Stata 文件、主键和验证合同

15.1 计划交付

DTA	Stata isid	用途
m2_product_treatment.dta	isid hs6 year	policy product panel
m2_connector_trade.dta	isid conn_iso3 hs6 year	product-level trade
m3_trade_sector_year.dta	isid conn_iso3 ana_sector year	sector-year trade
m3_trade_sector_window.dta	isid conn_iso3 ana_sector window_id	pre/shock/post
m4_production_long.dta	source-specific composite key	production evidence
m5_connector_evidence.dta	isid conn_iso3 ana_sector year	integrated interface
m6_connector_ranking.dta	isid conn_iso3 ana_sector rank_spec	ranking
m6_structural_moments.dta	isid moment_id scenario	model handoff

15.2 Storage types

- monetary levels：double；
- rates/shares/growth：double；
- year：int；
- month/quarter/flags：byte；
- counts：按范围使用 int 或 long；
- ISO3：str3；
- HS6：str6；
- sector/source/status：固定上限字符串，不使用易丢信息的过短类型。

15.3 必须运行的 Stata 检查

```
version 18
clear all
set more off

use "data/processed/stata/m3_trade_sector_year.dta", clear
describe
isid conn_iso3 ana_sector year
assert inrange(year, 2014, 2024)
assert cn_us_val >= 0 if !missing(cn_us_val)
assert cn_conn_val >= 0 if !missing(cn_conn_val)
assert ea_conn_val >= 0 if !missing(ea_conn_val)
assert conn_us_val >= 0 if !missing(conn_us_val)
assert inrange(sec_expo, 0, 1) if !missing(sec_expo)
assert qty_ok == 0 if !missing(qty_ok)

count
summarize cn_us_val cn_conn_val ea_conn_val conn_us_val sec_expo
display "DATA_HANDOFF_VALIDATION_SUCCESS"
```

最终 .do 要逐表检查 rows、keys、ranges、missing counts、totals 和 success marker，并把 log 写入项目 F 盘。

16. 每个变量完成后的登记模板

```
Canonical name:
Stata name:
Status:
Output file:
Grain and key:
Source fields:
Formula:
Window:
Unit:
Additivity:
Zero rule:
Missing rule:
Negative-value rule:
Suppression/confidentiality rule:
Denominator rule:
Small-base rule:
Winsorization/standardization:
Purpose:
Allowed interpretation:
Prohibited interpretation:
Validation checks:
Validation result:
Output hash:
Reviewer/date:
```

17. 当前施工顺序

1. M3.1 先只构造 connector-sector-year levels、coverage 和 sector treatment；
2. 验证 product-to-sector value reconciliation；
3. M3.2 再构造 window annual means；
4. M3.3 固定 zero/small-base/growth rules；
5. M3.4 构造 replacement、source input、exposure overlap 和 co-movement；
6. M3.5 保留所有 connectors/comparators；
7. M3.6 生成窗口、source bloc 和 exposure sensitivity；
8. M4 逐来源建立 production long；
9. M5 逐来源 left join 和 coverage；
10. M6 先 trade ranking，再 production grade，最后 structural moments；
11. M7 导出 Stata、运行 .do、冻结 release。

18. 版本记录

Version	Date	Change
v0.1.0	2026-07-27	建立 M2 已验证变量、M3-M6 planned variables、公式、用途、缺失规则、排名方案、structural moment 边界和 Stata handoff 合同。
v0.1.1	2026-07-27	与完整数据处理审计档案建立双向合同；规定 M3-M7 每步必须同步登记实际输出、validation、hash 和 Stata 接口。

下一次强制更新：M3.1 完成时，把本目录中本阶段涉及的 PLANNED_M3 变量更新为已构造状态，并补齐实际 output、hash、validation 和最终 Stata name；同时更新审计档案并重建两份 PDF。